

AJAY YADAV

Principal Engineer | 10+ Years Experience

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PROFESSIONAL SUMMARY

Principal Engineer with 10+ years of deep expertise in Node.js, TypeScript, AWS Serverless, and DevOps — specialized in designing and building scalable, cloud-native backend systems across fintech, logistics, and social impact domains. Proven track record architecting event-driven microservices, serverless infrastructures on AWS, and end-to-end CI/CD pipelines — delivering robust, high-performance applications from system design to production. Led engineering teams across 30+ microservices platforms, B2B data exchange systems, and payment processing engines serving thousands of transactions daily. Passionate about writing clean, reusable, and maintainable code, solving complex distributed systems challenges, and continuously improving scalability and system reliability. Track record of delivering high-impact platforms for global organizations including Google and Tata Trusts.

TECHNICAL SKILLS

Languages & Runtime: Node.js, JavaScript (ES6+), TypeScript

Backend & API: REST APIs, GraphQL, Express.js, Microservices, Serverless Architecture, Event-Driven Architecture, Webhooks, Streaming

Databases: MongoDB, DynamoDB, MySQL, PostgreSQL, Redis

Cloud — AWS: Lambda, Step Functions, SQS, SNS, Kinesis, API Gateway, EC2, S3, DynamoDB, CloudWatch, SAM, SES, IAM, KMS, Secrets Manager

Architecture & Design: System Design, Solution Architecture, Database Design, Cloud Infrastructure Design, API Design, Event-Driven Patterns, Queue-Based Architecture

Security & Auth: JWT, OAuth 2.0, RBAC, Bcrypt Hashing, End-to-End Encryption, OTP/2FA

Messaging: SMS, RCS, MMS, Twilio, AWS SNS

DevOps & Tooling: Git, GitLab, GitLab CI/CD Pipelines, AWS SAM, Serverless Framework, Docker

Project Management: Jira, Agile / Scrum, Sprint Planning, Backlog Management

Leadership & Process: Team Leadership, Mentoring, Code Review, Client Communication, Technical Documentation

PROFESSIONAL EXPERIENCE

Algoworks Technologies Private Limited

Oct 2019 – Present

Principal Engineer - Node JS

- Served as Principal Engineer across multiple concurrent enterprise products in fintech, logistics, and SaaS verticals — owning architecture, infrastructure, and delivery
- Worked directly with clients to translate business requirements into scalable system architectures and technical roadmaps
- Defined engineering standards, coding best practices, and architectural patterns adopted across backend teams
- Managed project delivery and sprint planning using Jira across multiple concurrent product teams
- Established and maintained GitLab CI/CD pipelines for automated testing, builds, and deployments across all projects
- Led and mentored teams of 4–6 engineers across SMSPayments, Trade Data Exchange, and ContractPlatformBuilder
- Owned end-to-end solution design — from requirements gathering and system design to deployment and production monitoring

Dhwani Rural Information System

Oct 2018 – Oct 2019

Senior Software Engineer

- Worked independently on all deliverables for high-impact social sector platforms (Google & Tata Trusts) with minimal supervision
- Contributed to product roadmap decisions alongside NGO and government program stakeholders

- Delivered production-grade features under tight field deployment deadlines serving thousands of rural workers across India
- Built survey and data collection systems handling real-time ingestion, accuracy validation, and large-scale reporting

DSign Infosystems Private Limited

Aug 2015 – Oct 2018

Software Engineer

- Built multiple client-facing projects from scratch, developing strong fundamentals in backend architecture and API design
- Worked directly with clients to gather requirements and deliver full-cycle software solutions
- Gained cross-stack expertise working across Node.js, MongoDB, MySQL, Redis, and AWS services
- Established foundation in scalable system design that shaped subsequent career growth into principal and architect roles

KEY PROJECTS

#01 SMSPayments · *Current · Active Development*

- Led a team of 4–6 backend engineers, owning end-to-end system architecture, tech stack selection, and AWS infrastructure design
- Designed and built a microservices architecture comprising 30+ independent services — covering payments, fraud, OTP, merchant onboarding, notifications, and reconciliation — enabling independent deployability and fault isolation
- Architected an event-driven payment engine processing thousands of transactions/day across SMS, RCS, and MMS messaging channels, using AWS SQS + SNS for message queuing, Kinesis for real-time event streaming, and Lambda for serverless processing
- Orchestrated complex transaction workflows using AWS Step Functions — managing multi-step payment, reconciliation, and fraud review state machines
- Designed fraud detection and risk scoring module to flag suspicious transactions in real time, monitored via CloudWatch alarms and dashboards
- Built OTP/2FA via SMS/RCS for secure transaction authorization using AWS SNS
- Engineered merchant onboarding workflows and bank/payment gateway integrations via REST APIs on API Gateway + EC2
- Implemented transaction reconciliation system ensuring data consistency across MongoDB and DynamoDB
- Deployed and managed entire serverless infrastructure using AWS SAM
- Secured sensitive payment credentials and API keys using AWS Secrets Manager, and enforced data encryption at rest using AWS KMS
- Improved system reliability and reduced latency through async, event-driven architecture and optimized service boundaries

Stack: Node.js, AWS Lambda, Step Functions, DynamoDB, SQS, SNS, Kinesis, API Gateway, CloudWatch, SAM, Serverless Framework, MongoDB, EC2, RCS, MMS, SMS, KMS, Secrets Manager

#02 Trade Data Exchange

- Led backend team and worked directly with client stakeholders to gather and translate business requirements into technical architecture
- Defined tech stack and infrastructure for a secure B2B data exchange platform between 2 enterprise clients
- Architected real-time data sync pipeline between enterprises using GraphQL APIs and webhook-based event notifications
- Built data transformation and normalization layer to reconcile differing data schemas across enterprise systems
- Optimized MongoDB and PostgreSQL aggregation pipelines for complex nested trade documents (invoices, bills of lading, shipment records)
- Managed secure storage of API credentials and encryption of sensitive trade data using AWS Secrets Manager and AWS KMS

Stack: Node.js, MongoDB, PostgreSQL, Redis, AWS S3, AWS EC2, AWS Lambda, GraphQL, REST APIs, Webhooks, AWS KMS, AWS Secrets Manager

#03 ContractPlatformBuilder

- Sole developer and architect — owned end-to-end system design, tech stack selection, and client requirements gathering for 2–10 small business clients
- Designed contract creation and templating engine supporting dynamic, reusable contract structures with field-level validation
- Implemented e-signature workflows with multi-role signing sequences and status tracking
- Built RBAC system controlling access for contract creators, reviewers, and signatories
- Developed automated notifications and reminders for pending signatures and contract deadlines via email (AWS SES), SMS (AWS SNS), and WhatsApp (Twilio)

Stack: Node.js, MongoDB, MySQL, Redis, AWS S3, AWS SES, AWS SNS, Twilio, JWT

#04 InternetSaathi · Google & Tata Trusts

- Built core modules and owned data accuracy & quality for a national-scale survey platform serving 30,000+ field workers across rural India, influencing 12 million women
- Solved high-concurrency challenges handling simultaneous survey submissions from thousands of active field workers
- Designed multi-tenant survey management system supporting multiple independent client campaigns on a shared platform
- Built reporting and analytics module for clients including Niti Aayog, delivering insights on digital literacy program outcomes
- Implemented data accuracy and deduplication pipelines to ensure integrity of survey responses at scale

Stack: Node.js, Express.js, MongoDB, MySQL, Redis, REST APIs, JWT, AWS EC2, CSV Processing, Aggregation Pipelines, Caching

#05 DELTA · Tata Trusts

- Managed functional requirements and built core data modules for a real-time mobile survey platform serving thousands of field workers across India
- Engineered real-time data ingestion pipeline from mobile devices into MongoDB for live program monitoring
- Built ad-hoc reporting and analytics layer for Tata Trusts program officers to track government scheme convergence
- Handled large nested dataset challenges — designed schema and aggregation strategies for complex hierarchical survey data
- Implemented CSV streaming for bulk data exports enabling offline analysis by program teams
- Optimized MongoDB queries for report generation across high-volume, deeply nested collections

Stack: Node.js, Express.js, MongoDB, MySQL, Redis, REST APIs, JWT, AWS EC2, CSV Processing, Aggregation Pipelines, Caching

EDUCATION

Bachelor of Technology (B.Tech), Honours — CGPA: 8.08 / 10

2011 – 2015

NRI Engineering College, Gwalior

ADDITIONAL INFORMATION

Total Experience: 10+ Years

Languages: English (Professional), Hindi (Native)

Availability: Open to full-time, contract, and remote opportunities